Experiment No. 5

Experiment Name

Feature Extraction and Image Analysis using SIFT and HOG Descriptors

Aim

To implement Scale-Invariant Feature Transform (SIFT) and Histogram of Oriented Gradients (HOG) techniques for feature extraction and analyze their effectiveness in image representation, object recognition, and computer vision applications.

Purpose

Feature extraction is a fundamental step in computer vision that transforms raw image data into meaningful descriptors suitable for object recognition, image matching, tracking, and classification. Robust feature descriptors should be invariant to changes in scale, rotation, illumination, and viewpoint.

In this experiment, students will implement SIFT and HOG feature extraction techniques on real-world images and compare their performance in terms of feature detection, robustness, computational complexity, and suitability for different computer vision applications.

Prerequisites

- Basic knowledge of Digital Image Processing
- Familiarity with Python programming
- Python libraries: OpenCV (cv2), scikit-image, NumPy, and Matplotlib
- Understanding of image gradients, keypoints, and feature descriptors
- Jupyter Notebook, Google Colab, or Visual Studio Code

Steps

1. Import the required libraries and load a real-world image or image dataset for analysis.
2. Convert the image into grayscale, if required, and perform basic preprocessing to improve image quality.
3. Apply the SIFT (Scale-Invariant Feature Transform) algorithm to detect keypoints and compute feature descriptors.
4. Visualize the detected keypoints on the image and analyze their distribution across different image regions.

5. Extract Histogram of Oriented Gradients (HOG) features from the same image using appropriate parameters such as cell size, block size, and orientation bins.
6. Visualize the HOG descriptor to understand the gradient orientation patterns captured from the image.
7. Compare the characteristics of SIFT keypoints and HOG descriptors with respect to scale invariance, rotation invariance, and computational efficiency.
8. Apply SIFT or HOG features to perform basic image matching or object recognition using two similar images.
9. Evaluate the strengths and limitations of both feature extraction techniques for different computer vision tasks.
10. Document the observations and discuss suitable applications where SIFT and HOG descriptors can be effectively utilized.

Questions

1. What is feature extraction, and why is it important in computer vision?
2. Explain the working principle of Scale-Invariant Feature Transform (SIFT).
3. What are keypoints and feature descriptors in image analysis?
4. Explain the concept of Histogram of Oriented Gradients (HOG) and its significance.
5. Compare SIFT and HOG based on robustness, computational complexity, and practical applications.
6. Why is SIFT considered invariant to scale and rotation?
7. Mention three real-world applications where HOG descriptors are commonly used.
8. Why is feature extraction performed before image classification or object detection?
9. What are the advantages and limitations of handcrafted feature descriptors compared to deep learning-based feature extraction?
10. How do feature extraction techniques contribute to image matching, face recognition, and object detection systems?